# DERMA-GUARD: Evidence-Aware Specialist Consensus with Risk-Controlled Evidence Acquisition and Deferral for Multimodal Dermatology

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***Abstract*—Skin-lesion classifiers rarely fail for lack of accuracy on clean benchmarks. They fail because they answer every case with the same apparent confidence: a blurred photograph, missing history, or a photograph and a history that point to different diagnoses all produce a label. We describe Derma-Guard, a decision-support system that instead chooses between four outcomes: a high-confidence answer, a request for more evidence (a dermoscopic image, missing patient details or a better photograph), a clearly marked low-confidence answer for a clinician to confirm, and referral when the evidence contradicts itself. Six specialists (evidential heads for the photograph, dermoscopy and metadata, an external legacy model, a nearest-neighbour vote and a class-balanced embedding classifier) are profiled case by case by an evidence manager, and a learned gate decides how much each should count; the gate was chosen from seven combination rules under five-fold cross-fitting. A calibrated, cost-aware policy with a Learn-then-Test bound on missed cancers makes the final choice. On 787 held-out MILK10k lesions, 89.1% of patients receive an answer and 57.1% a high-confidence one that is 77.1% accurate over eleven diagnoses and misses 3.1% of cancers, against 4.9% for a learned rejector ($p = 0.011$) and 3.8% for confidence thresholding at the same coverage. Across four fusion back-ends the policy's miss rate stays between 3.0% and 3.5%, while thresholding ranges from 3.5% to 9.3%: abstention by confidence is only as safe as the probabilities beneath it. Controlled image damage shows that evidential uncertainty misses blur, darkness, noise and occlusion; explicit checks remove every confident answer on damaged input. Without retraining, the system stays within its safety target on PAD-UFES-20 and Derm7pt, while its fine-grained diagnosis does not transfer. Re-evaluating our earlier prototype, whose reported accuracies of 84–90% were not reproducible, places its models at 57–75% on unseen patients.***

***Index Terms*—dermatology, multimodal fusion, evidential deep learning, selective prediction, conformal prediction, risk control, evidence acquisition, clinical decision support**

## I. Introduction

A patient who photographs a mole and asks an app whether it is dangerous gets an answer almost every time. The answer comes with a number, often above 90%, and that number looks the same whether the photograph was sharp or blurred, whether the patient's age and the lesion's location agree with the image, and whether the lesion resembles anything in the training data. Deep networks reached dermatologist-level accuracy on curated images years ago [1], so accuracy on benchmarks is no longer the main obstacle to safe use. Behaviour is.

A careful dermatologist behaves differently. When the evidence is clear they give a diagnosis. When it is not, they look more closely with a dermatoscope, give a provisional opinion and ask for confirmation, or refer the patient when the history and the lesion tell different stories. We wanted to know whether a model can be built to behave this way, and how much it costs.

This paper is also the second attempt. The first version of Derma-Guard already had the right intuitions (a panel of specialist models, a sceptical component that questions their evidence, and trust that depends on the suspected diagnosis), but its weights were typed in by hand. When we audited it, its reported accuracies did not survive: the split was made by photograph rather than by patient, and its evaluation code adjusted predictions toward preset accuracy targets (Section V-E). We kept the ideas, replaced every hand-set number with one fitted on validation data, and made every result a script that can be re-run.

Our two base papers (Section II) are HOT AI [2], the closest task, which asks for a dermoscopic image when the photograph is not enough, and Learn-then-Test [3], which provides the statistical guarantee. We compare against both, or against a faithful baseline built from them, on the same probabilities.

**Contributions.** (i) A six-specialist team whose per-case weights are learned from all of the evidence manager's signals, chosen from seven rules by cross-fitted validation, with an exhaustive subset study and an oracle ceiling. (ii) A calibrated four-outcome policy that can request three kinds of missing evidence and certifies the missed-cancer rate of its confident answers. (iii) The finding that confidence-based abstention is only as safe as the fusion beneath it, whereas the calibrated policy is stable across fusion back-ends. (iv) A controlled degradation benchmark, two external tests without retraining, and an honest re-evaluation of the earlier prototype's seven models. (v) A working care platform, with every system variant available side by side in its interface.

## II. Base Papers and Related Work

**Base paper 1: HOT AI** [2] is the closest task. From a clinical image it produces a hierarchical diagnosis, an out-of-distribution alert and a recommendation to take a dermoscopic image when the photograph alone is not enough, and then fuses both images. It establishes that an AI which asks for dermoscopy is not new. What Derma-Guard adds is an explicit cost for acquiring and for deferring, a separate defer action for contradictory evidence, a certified bound on missed cancers, and stress tests on a public paired dataset. Because HOT AI's model and data are not public, we compare against a HOT-style rule built from it: ask the most uncertain patients for dermoscopy, at the same rate as our policy, on the same probabilities.

**Base paper 2: Learn-then-Test** [3] chooses decision thresholds on held-out data so that a chosen risk is bounded with a finite-sample guarantee when future cases resemble the calibration cases. We use it unchanged for the risk that matters here, the share of answered cancers called benign, inside a three-action policy, and we measure what happens to the guarantee on two other datasets.

**Data and evaluation.** MILK10k [4] pairs a clinical close-up and a dermoscopic image for every lesion; a reader study on it [5] found that dermoscopy has higher sensitivity for malignancy and the close-up higher specificity, each adding value, which is the clinical reason to ask for dermoscopy selectively. JI-ADF [6] reports multimodal fusion on MILK10k's hidden test, which we did not use, so its numbers are not comparable with ours. We evaluate abstention with AUGRC [7] and matched-coverage comparisons, and acquisition against random and oracle bounds in the spirit of CAMA [8].

**Classification and fusion.** Deep networks reached dermatologist level on curated images [1], typically on residual networks [9]; dermatology foundation models [10] are the frontier. Multimodal work fuses images with metadata [11] or pairs clinical and dermoscopic images [12]; PAD-UFES-20 [13] provides smartphone photographs with metadata. These models improve the prediction that a policy sits on; fusion is a component here, not the contribution.

**Uncertainty, conformal sets and deferral.** Evidential deep learning [14] and Dempster–Shafer fusion [15], [16] give per-case ignorance and conflict; temperature scaling [17], isotonic regression [18] and Mahalanobis scores [19] calibrate and detect unusual inputs. Conformal prediction has been validated for skin lesions [20], made group-conditional by skin type [21], and is smallest on average in its least-ambiguous form [22]. Chow's rule [23], selective classification [24] and learning to defer [25] decide when not to answer. Each ingredient exists; the open question is whether combining a cost model, a statistical guarantee, a three-action policy and combined-failure testing on public paired data measurably helps.

## III. Data and Evaluation Protocol

MILK10k [4] has 5,240 lesions, each with one clinical close-up and one dermoscopic image, eleven diagnoses (95.7% confirmed by histopathology; 71.7% malignant), age, sex, body site, a skin-tone class and seven automatically computed concept scores. Without patient identifiers, we split by lesion into 3,667 training, 786 validation and 787 test lesions, asserting that no lesion appears twice. Validation fits everything that is fitted after the network: fusion rules, reliabilities, calibration, conformal quantiles, the risk threshold, the acquisition estimator and every choice between methods. The test split was scored once. The data are heavily imbalanced: basal cell carcinoma makes up 48% of lesions, while four classes have fewer than ten test lesions each, which is why we report macro-F1 and balanced accuracy next to accuracy throughout.

TABLE I
The two base papers (top) and the other methods we compare against, with what we measured (MILK10k test, $c_{\text{miss}} = 5$).

| Work | Its contribution | Our use and measured result |
|---|---|---|
| HOT AI [2] | asks for dermoscopy when the photo is insufficient; OOD alert | adds costs, defer action, certified bound; HOT-style "ask the most uncertain" rule: acquisition efficiency $+0.26$ vs $+0.07$ for our estimator (it wins), but answering everyone afterwards misses 5.6% vs 3.1% for our confident answers |
| Learn-then-Test [3] | risk-controlling thresholds | certified at both stages ($\hat{\lambda}$ 0.218, 0.276); held on PAD-UFES-20 (0.0%) and Derm7pt (3.8%) |
| Learning to defer [25] | learned deferral | L2D-style rejector misses 4.9% vs our 3.1% ($p = 0.011$) |
| Confidence threshold [24] | selective classification | 3.8% vs 3.1% (n.s.); 3.5–9.3% across four fusions |
| AUGRC [7] | abstention metric | ignorance 0.0820 vs softmax 0.0757 (softmax better) |
| MILK10k [4], [5] | paired data; each modality adds value | dermoscopy head AUROC 0.854 vs photo 0.787; fused 0.885 |

**External data.** PAD-UFES-20 [13] (2,298 smartphone photographs, six classes, no dermoscopy) and Derm7pt [12] (1,003 lesions with both images after mapping its diagnoses onto MILK10k classes) are used only for testing, without retraining or recalibration.

**Metrics.** For a system that may abstain we report coverage (share of high-confidence answers), accuracy among them and the missed-cancer rate (share of answered malignant lesions called benign), always comparing systems at equal coverage; macro-F1 and balanced accuracy for the rare classes; malignant-vs-benign AUROC; AUGRC [7]; acquisition efficiency

$$\mathrm{AE} = \frac{G_\pi - G_{\text{rand}}}{G_{\text{oracle}} - G_{\text{rand}}} \tag{1}$$

for the choice of whom to send for dermoscopy; and 95% bootstrap intervals with paired tests (2,000 resamples).

## IV. Method

Fig. 1 shows the pipeline. A case moves left to right: specialists give opinions, the evidence manager profiles them, a fusion rule turns the profile into weights, and the policy decides.

### A. Specialists

**S1–S3: evidential heads.** An ImageNet ResNet50 [9] shared by both image types and a residual metadata network

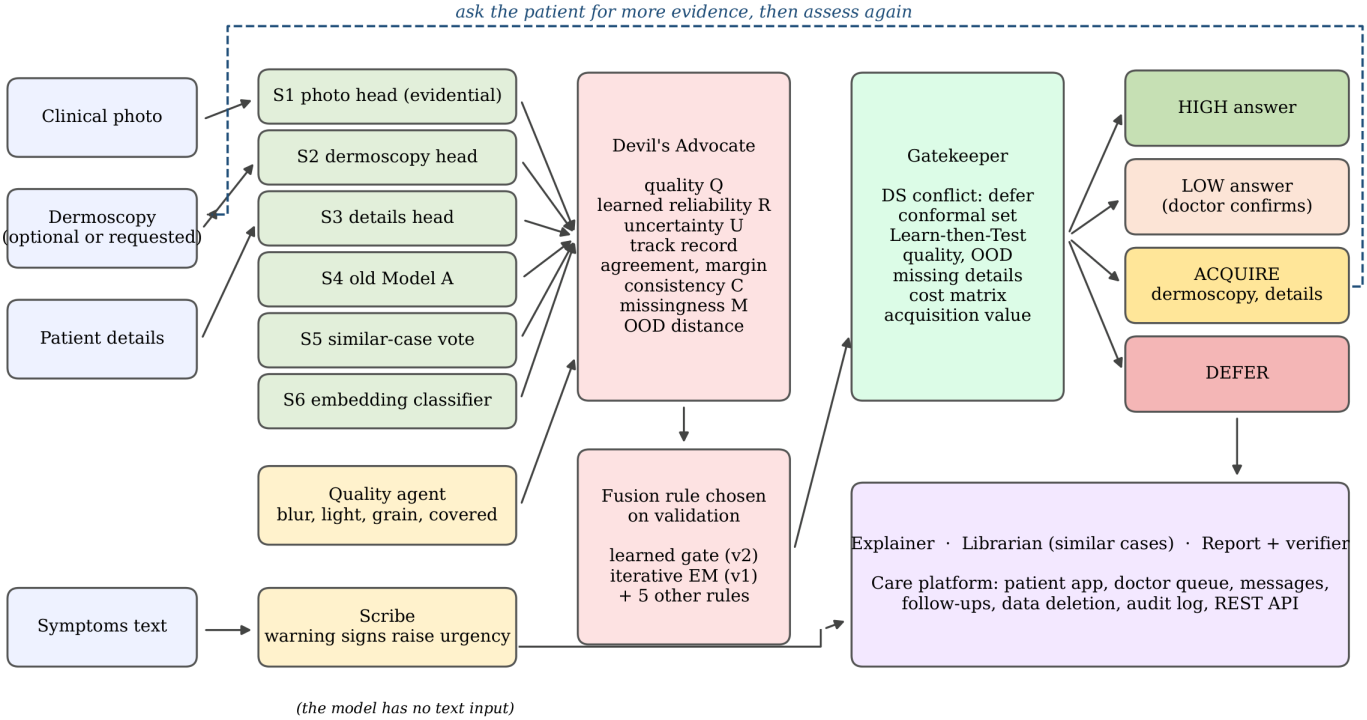



Fig. 1. The Derma-Guard pipeline: specialists, evidence manager, fusion and the four-outcome policy.

produce three 256-dimensional embeddings, each feeding an evidential head with non-negative evidence $e_m$ [14]. With $\alpha_m = e_m + 1$ and $S_m = \sum_k \alpha_{m,k}$, belief and ignorance are $b_{m,k} = e_{m,k}/S_m$ and $u_m = K/S_m$; a head contributes its pignistic distribution $p_m = b_m + u_m/K$. The heads are also combined by Dempster's rule [16],

$$C = \sum_{i \neq j} b_{1,i} b_{2,j}, \qquad u = \frac{u_1 u_2}{1 - C},$$
$$b_k = \frac{b_{1,k} b_{2,k} + b_{1,k} u_2 + b_{2,k} u_1}{1 - C} \tag{2}$$

which leaves a missing modality ($u = 1$) exactly without effect and separates contradiction ($C$) from shared ignorance ($u$). The network was trained on a Kaggle GPU for 25 epochs with modality dropout 0.25, so that it learns to say "I don't know" for an absent input.

**S4: legacy model.** The earlier prototype's image model, trained on PAD-UFES-20, is applied unchanged; its six classes are mapped to MILK10k and mixed with 10% uniform mass so unsupported classes are unknown rather than impossible.

**S5, S6: case-based specialists.** S5 is a vote of the 15 most similar training lesions in embedding space, weighted by $\exp(5\,\mathrm{sim})$. S6 is a multinomial logistic regression over all training embeddings with class weights inversely proportional to frequency, so that rare diagnoses are not drowned out; its regularisation was chosen on validation.

### B. The evidence manager

For specialist $m$ on case $x$ the evidence manager computes ten signals: quality $Q_m$ (share of image statistics inside the training 5th–95th percentiles), a reliability $R_m$ (logistic regression fitted on validation predicting whether $m$ is right), normalised entropy $U_m$, confidence, margin, agreement with the other specialists, the measured track record $\pi_m(\hat{y}_0) = \Pr(m \text{ correct} \mid y = \hat{y}_0)$ at the provisional diagnosis, the team's Jensen–Shannon consistency, missingness, and the Mahalanobis out-of-distribution distance [19]. The earlier prototype multiplied hand-set versions, $E_m = Q_m R_m U_m C S M_m$, and normalised them into weights; because consistency $C$ and sufficiency $S$ are equal for all specialists of a case, they cancelled, and only three signals ever moved the weights.

### C. Fusion rules and how one is chosen

**Iterative diagnosis-adaptive fusion (v1)** keeps the prototype's EM-style loop [26]: weights $w_m \propto (Q_m R_m)^a$ give a provisional diagnosis $\hat{y}$; then up to three times

$$w_m = \mathrm{softmax}_m\big(a \log Q_m R_m + b \log \pi_m(\hat{y}) + g \log \kappa_m\big),$$

with $\kappa_m = \rho$ for specialists disagreeing with $\hat{y}$, and the diagnosis is updated.

**The learned gate (v2)** lets every signal act. With the standardised signal vector $\phi_m(x)$,

$$w_m(x) = \underset{m \in \text{present}}{\mathrm{softmax}}\big(\beta_m + \theta_m^\top \phi_m(x)\big), \tag{3}$$

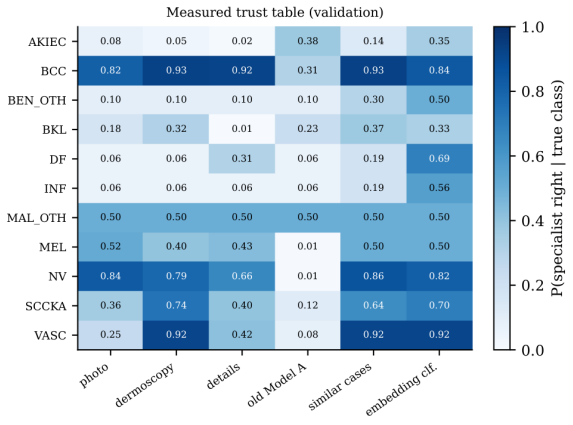



Fig. 2. Measured trust table: probability that each specialist is right given the true class (validation).

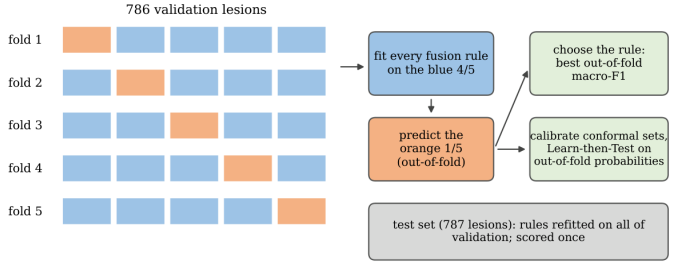



Fig. 3. Cross-fitting protocol on the validation lesions.

and the opinions are pooled either as a mixture $\sum_m w_m p_m$ or as a product of experts $\propto \prod_m p_m^{w_m}$, followed by temperature scaling. $\theta$, $\beta$ and the temperature minimise validation log-loss with an $\ell_2$ penalty.

Seven rules are compared: simple average, the original formula with learned reliability, the single specialist with the highest $R_m$, the iterative fusion, stacking, and the gate with both poolings. Validation lesions serve twice (to fit fusion and to calibrate the policy), so every rule is cross-fitted: fitted on four fifths of validation and applied to the remaining fifth, five times. The out-of-fold probabilities are used both to select the rule (best macro-F1, fixed before any test result was seen) and to calibrate the policy. We also refit the selected rule on all 31 (stage A) and 63 (stage B) specialist subsets, and report an oracle that picks, per case, the specialist giving the true class most mass. Fig. 3 summarises the protocol.

### D. The decision policy

Stage A uses the photograph and metadata; stage B adds dermoscopy. The stated diagnosis is the most probable class; costs are 0 for a correct answer, 1 within the benign or malignant group, $c_{\text{miss}} = 5$ for a cancer called benign and 2 for the reverse, with 0.5 to acquire dermoscopy and 1.5 for a clinician review. Gates are: Dempster–Shafer conflict above its validation 90th percentile (Defer); a Mondrian least-ambiguous conformal set over {benign, malignant} ($\alpha = 0.1$) [21], [22] that still contains "malignant" when the answer would be benign; a Learn-then-Test threshold [3] on $1 - u$ such that $\Pr(\text{missed-cancer rate} \le 0.05) \ge 0.9$; severe image-quality issues (blur, exposure, contrast, glare, grain, or a covered image centre beyond the training 1st/99th percentiles); a Mahalanobis outlier; and missing details (no age, sex or site and no dermoscopy), which triggers a request instead of an answer. If no gate fires and answering is cheapest, the answer is High. Otherwise the system requests dermoscopy when the expected cost after acquisition, estimated by a gradient-boosted regressor fitted on validation lesions whose dermoscopy is known, plus 0.5, is below the cost of review; else it gives a Low answer for a clinician to confirm. Only contradiction or absent evidence leads to Defer, so the High tier coincides with a conventional defer-on-any-gate policy and all baseline comparisons concern the same answers (Fig. 4). An earlier version deferred on every failed check, which sent every patient without a confident answer to a clinician with nothing at all; the Low tier keeps those patients informed while still requiring confirmation.

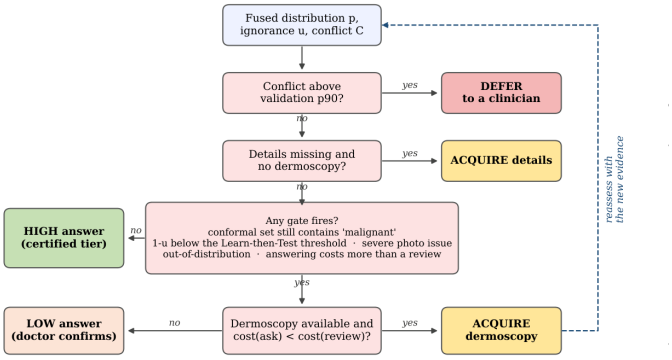



Fig. 4. The decision policy and its four outcomes.

### E. Agents and platform

Around the policy, a quality agent advises on retakes, a scribe raises triage urgency when the patient's own words mention warning signs (with negation handling; it never changes the diagnosis), a librarian shows similar confirmed training lesions, an explainer states who said what, why dissenting specialists disagree and where the weight went, and a report writer may use a local language model whose output is discarded if a verifier finds any diagnosis or percentage not in the evidence. A care platform built on SQLite routes each outcome: a confident answer schedules a follow-up, a request moves the case to "awaiting patient" until the evidence arrives, and Low answers and deferrals enter a clinician queue ordered by kind and urgency. A follow-up photograph whose probability of malignancy rises by 0.15 or more returns the case as urgent. Every assessment stores the policy version that produced it, every event goes to an append-only audit log,

TABLE II
THE SEVEN MODELS OF THE BENCHMARK.

| Code | Name | Encoders |
|---|---|---|
| A | Image only | ResNet50 + multi-head self-attention |
| B | Metadata only | Residual MetaBlock MLP |
| C | Free text only | Bio_ClinicalBERT |
| D | Image + Metadata | ViT + MetaBlock MLP |
| E | Image + Free text | ViT + Bio_ClinicalBERT |
| F | Free text + Metadata | Bio_ClinicalBERT + MetaBlock MLP |
| G | Full multimodal | ViT + MetaBlock MLP + Bio_ClinicalBERT |

and patients can delete a case or their whole record, including uploaded photographs. A Streamlit application exposes patient, clinician, stress-testing and research views, and a "compare systems" view runs any held-out lesion through every specialist, every fusion rule and every baseline side by side; a FastAPI service exposes the same workflow. The code base has 230 automated tests.

For one held-out melanoma in our test set, for instance, the explainer reports that the photograph head named melanoma but was only 10% sure, that the legacy model said benign keratosis because it "was trained on a different dataset and cannot name five of the eleven diagnoses" and was right on only 1% of validation melanomas, and that most of the weight went to the metadata head, the embedding classifier and the photograph head, which agreed with the team and had high learned reliability for this case. The system then asked for the dermoscopic image. Every number in such an explanation is read from the evidence profile; the language model, when used, may only reword it.

## V. RESULTS

### A. Specialists and fusion

The seven benchmark models (A image, B metadata, C free text, D image + metadata, E image + text, F text + metadata, G all three; Table II) were scored on the 459-image test partition of PAD-UFES-20 (169 BCC, 146 ACK, 49 NEV, 47 SEK, 38 SCC, 10 MEL), using the same learned softmax-normalised fusion weights for the fused models. Table III and Figs. 5 and 6 summarise them. Model E (image + text) is the most accurate, with 89.98% (413 of 459), the highest macro precision (0.8874), specificity (0.9771), F1-score (0.8926) and AUC (0.9982). Model G (all three inputs) is second with 87.80%, then B and D with 86.27% each, F with 85.40%, C with 85.19% and A with 83.66%. Macro sensitivity is highest for D (0.9332), followed by G (0.9247) and A (0.9198), and macro AUC is above 0.994 for every model.

Every added input helps in most comparisons (Table IV). Metadata raises the image model by 2.61 points (A to D) and free text by 6.32 points (A to E); the image raises the text model by 4.79 points (C to E). Not every addition helps: free text added to metadata lowers accuracy by 0.87 points (B to F), and metadata added to image and text lowers it by 2.18 points (E to G), although it raises ACK and SCC recall. Free text and metadata together (F, 85.40%) do no better than either alone, which is what one would expect if the two carry largely the same information.

TABLE III
OVERALL TEST PERFORMANCE (MACRO AVERAGES UNLESS STATED).

| Model | Acc. (%) | Prec. | Sens. | Spec. | F1 | AUC | AUPRC |
|---|---|---|---|---|---|---|---|
| A | 83.66 | 0.7878 | 0.9198 | 0.9650 | 0.8314 | 0.9952 | 0.9906 |
| B | 86.27 | 0.8291 | 0.8583 | 0.9681 | 0.8365 | 0.9954 | 0.9665 |
| C | 85.19 | 0.8004 | 0.9017 | 0.9674 | 0.8354 | 0.9949 | 0.9826 |
| D | 86.27 | 0.8124 | 0.9332 | 0.9706 | 0.8545 | 0.9967 | 0.9935 |
| E | 89.98 | 0.8874 | 0.9056 | 0.9771 | 0.8926 | 0.9982 | 0.9848 |
| F | 85.40 | 0.8029 | 0.8959 | 0.9677 | 0.8358 | 0.9948 | 0.9801 |
| G | 87.80 | 0.8270 | 0.9247 | 0.9734 | 0.8618 | 0.9967 | 0.9895 |

TABLE IV
EFFECT OF ADDING A MODALITY (CHANGE IN CORRECT TEST IMAGES AND IN ACCURACY POINTS).

| Comparison | Images | Points | Modality added |
|---|---|---|---|
| D vs A | +12 | +2.61 | metadata to image |
| E vs A | +29 | +6.32 | free text to image |
| G vs A | +19 | +4.14 | metadata and free text to image |
| D vs B | +0 | +0.00 | image to metadata |
| F vs B | -4 | -0.87 | free text to metadata |
| G vs B | +7 | +1.53 | image and free text to metadata |
| E vs C | +22 | +4.79 | image to free text |
| F vs C | +1 | +0.21 | metadata to free text |
| G vs C | +12 | +2.61 | image and metadata to free text |
| G vs D | +7 | +1.53 | free text to image + metadata |
| G vs E | -10 | -2.18 | metadata to image + free text |
| G vs F | +11 | +2.40 | image to free text + metadata |

TABLE V
CORRECT PREDICTIONS AND WILSON 95% INTERVALS FOR ACCURACY.

| Model | Correct | Errors | Acc. (%) | Wilson 95% CI (%) |
|---|---|---|---|---|
| A | 384/459 | 75 | 83.66 | [80.0, 86.8] |
| B | 396/459 | 63 | 86.27 | [82.8, 89.1] |
| C | 391/459 | 68 | 85.19 | [81.6, 88.1] |
| D | 396/459 | 63 | 86.27 | [82.8, 89.1] |
| E | 413/459 | 46 | 89.98 | [86.9, 92.4] |
| F | 392/459 | 67 | 85.40 | [81.9, 88.3] |
| G | 403/459 | 56 | 87.80 | [84.5, 90.5] |

The Wilson 95% intervals (Table V) are about 6–7 points wide; only those of E (86.9–92.4%) and A (80.0–86.8%) fail to overlap, and then barely. Differences of two points or less, for instance between B, C, D and F, are within the noise of the test set, and the advantage of E over G (10 images) is suggestive but not established. Macro AUPRC ranges from 0.9665 (B) to 0.9935 (D).

The gate-learned fusion weights give the image the largest weight whenever it is present (Table VI): 0.786 against 0.214 for metadata in D and 0.825 against 0.175 for free text in E. In the full model G the weights are 0.717, 0.178 and 0.104, so free text receives the smallest weight of any branch in any model.

### B. Per class

Fig. 7 and Tables VII and VIII give the confusion matrices and the per-class recall and precision. In all seven matrices the rows for NEV, SEK and MEL lie on the diagonal: every error is a true BCC, ACK or SCC lesion assigned to a wrong class, and the team never calls a cancer a nevus or a seborrheic

**DERMA-GUARD MULTIMODAL COMPREHENSIVE BENCHMARK SCORECARD**

*Dataset: PAD-UFES-20 (Total N = 2,298 | Train N = 1,953 [85%] | Val N = 229 [10%] | Test N = 459 [20%])*

| Code | Model Name | Modality Paradigm | Accuracy | Precision (M) | Sensitivity (M) | Specificity (M) | F1-Score (M) | AUC (M) | AUPRC (M) | TP | TN | FP | FN | Dataset Split (Train/Val/Test) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Model A | Model A (Only Image) | Image (ResNet50 + MHSA ViT) | 83.66% | 0.7878 | 0.9198 | 0.9650 | 0.8314 | 0.9952 | 0.9906 | 384 | 2220 | 75 | 75 | Total=2298 (1953/229/459) |
| Model B | Model B (Only Metadata) | Metadata (Residual MetaBlock MLP) | 86.27% | 0.8291 | 0.8583 | 0.9681 | 0.8365 | 0.9954 | 0.9665 | 396 | 2232 | 63 | 63 | Total=2298 (1953/229/459) |
| Model C | Model C (Only Free Text) | Free Text (Bio_ClinicalBERT) | 85.19% | 0.8004 | 0.9017 | 0.9674 | 0.8354 | 0.9949 | 0.9826 | 391 | 2227 | 68 | 68 | Total=2298 (1953/229/459) |
| Model D | Model D (Image + Metadata) | Image + Metadata (ViT + MetaBlock MLP) | 86.27% | 0.8124 | 0.9332 | 0.9706 | 0.8545 | 0.9967 | 0.9935 | 396 | 2232 | 63 | 63 | Total=2298 (1953/229/459) |
| Model E | Model E (Image + Free Text) | Image + Free Text (ViT + Bio_ClinicalBERT) | 89.98% | 0.8874 | 0.9056 | 0.9771 | 0.8926 | 0.9982 | 0.9848 | 413 | 2249 | 46 | 46 | Total=2298 (1953/229/459) |
| Model F | Model F (Free Text + Metadata) | Free Text + Metadata (Bio_ClinicalBERT + MetaBlock MLP | 85.40% | 0.8029 | 0.8959 | 0.9677 | 0.8358 | 0.9948 | 0.9801 | 392 | 2228 | 67 | 67 | Total=2298 (1953/229/459) |
| Model G | Model G (Full Multimodal) | Full Multimodal (ViT + MetaBlock MLP + Bio_ClinicalBERT | 87.80% | 0.8270 | 0.9247 | 0.9734 | 0.8618 | 0.9967 | 0.9895 | 403 | 2239 | 56 | 56 | Total=2298 (1953/229/459) |

Fig. 5. Benchmark scorecard of Models A to G with the 1,953 / 229 / 459 dataset split.

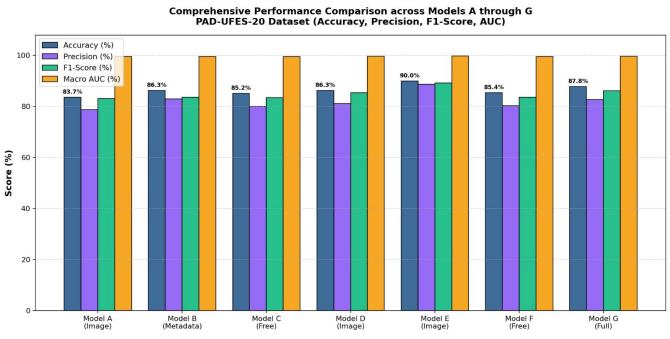



Fig. 6. Accuracy, precision, F1-score and macro AUC of Models A to G.

TABLE VI
MEAN LEARNED FUSION WEIGHTS OF THE FUSED MODELS.

| Model | First weight | Second weight | Third weight |
|---|---|---|---|
| D | 0.7863 (image) | 0.2137 (metadata) | – |
| E | 0.8252 (image) | 0.1748 (free text) | – |
| F | 0.2438 (free text) | 0.7562 (metadata) | – |
| G | 0.7172 (image) | 0.1785 (metadata) | 0.1044 (free text) |

TABLE VII
PER-CLASS SENSITIVITY (RECALL, %).

| Model | BCC | ACK | NEV | SEK | SCC | MEL |
|---|---|---|---|---|---|---|
| A | 79.3 | 72.6 | 100.0 | 100.0 | 100.0 | 100.0 |
| B | 94.1 | 78.8 | 100.0 | 100.0 | 42.1 | 100.0 |
| C | 79.9 | 82.2 | 100.0 | 100.0 | 78.9 | 100.0 |
| D | 80.5 | 79.5 | 100.0 | 100.0 | 100.0 | 100.0 |
| E | 98.8 | 78.8 | 100.0 | 100.0 | 65.8 | 100.0 |
| F | 81.7 | 82.2 | 100.0 | 100.0 | 73.7 | 100.0 |
| G | 81.7 | 86.3 | 100.0 | 100.0 | 86.8 | 100.0 |

TABLE VIII
PER-CLASS PRECISION (%).

| Model | BCC | ACK | NEV | SEK | SCC | MEL |
|---|---|---|---|---|---|---|
| A | 89.3 | 84.8 | 83.1 | 100.0 | 65.5 | 50.0 |
| B | 82.4 | 95.0 | 86.0 | 100.0 | 57.1 | 76.9 |
| C | 88.2 | 86.3 | 87.5 | 100.0 | 68.2 | 50.0 |
| D | 91.9 | 86.6 | 86.0 | 100.0 | 70.4 | 52.6 |
| E | 87.0 | 99.1 | 86.0 | 100.0 | 69.4 | 90.9 |
| F | 87.3 | 87.6 | 87.5 | 100.0 | 66.7 | 52.6 |
| G | 91.4 | 88.1 | 90.7 | 100.0 | 73.3 | 52.6 |

TABLE IX
CANCERS (BCC, SCC, MEL) KEPT IN A CANCER CLASS AND NON-CANCERS (ACK, NEV, SEK) KEPT IN A NON-CANCER CLASS.

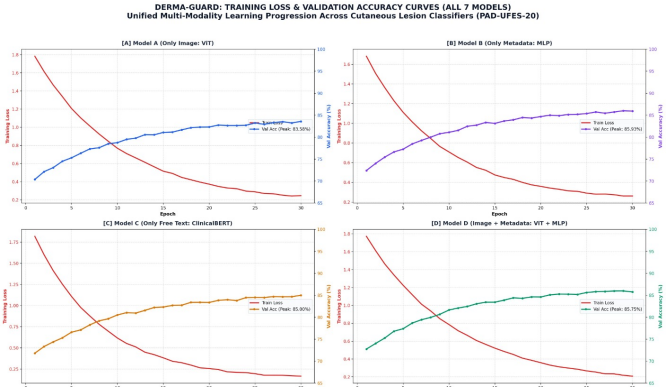
| Model | Cancers kept | % | Non-cancers kept | % |
|---|---|---|---|---|
| A | 198 / 217 | 91.2 | 212 / 242 | 87.6 |
| B | 211 / 217 | 97.2 | 219 / 242 | 90.5 |
| C | 198 / 217 | 91.2 | 223 / 242 | 92.1 |
| D | 199 / 217 | 91.7 | 220 / 242 | 90.9 |
| E | 216 / 217 | 99.5 | 219 / 242 | 90.5 |
| F | 200 / 217 | 92.2 | 223 / 242 | 92.1 |
| G | 200 / 217 | 92.2 | 227 / 242 | 93.8 |

keratosis. BCC is where the models differ most in recall, from 79.3% (A) and 79.9% (C) to 94.1% (B) and 98.8% (E). ACK has the lowest recall of the two large classes in every model (72.6% for A to 86.3% for G). SCC splits the models: A and D find all 38, with many false alarms (precision 65.5% and 70.4%), whereas B (recall 42.1%) and E (65.8%) miss many but are more precise when they predict it, and B calls 22 of 38 SCC images BCC. Melanoma precision is 90.9% for E and 76.9% for B but only 50.0–52.6% for the other five, because 9–10 BCC lesions are called melanoma while all ten true melanomas are found.

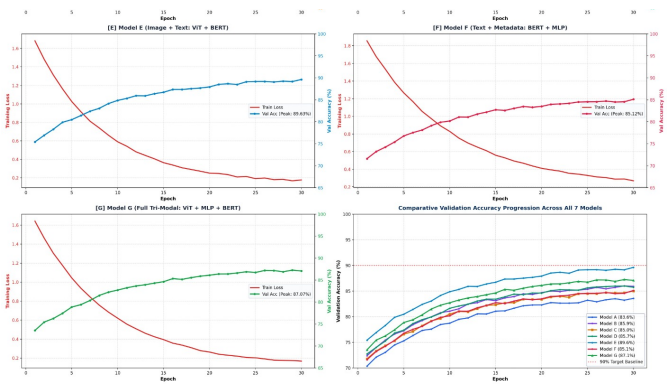
### C. Decisions

Grouping BCC, SCC and MEL as cancers (217 images) against ACK, NEV and SEK (242 images), E keeps 216 of 217 cancers in a cancer class (99.5%) and B 211 (97.2%), against 198–200 (91.2–92.2%) for A, C, D, F and G; G has the fewest non-cancers pushed into a cancer class (227 of 242 kept, 93.8%). Under the stricter high-risk grouping (BCC, ACK, SCC, MEL against NEV, SEK) no model called a benign lesion high-risk, and the only high-risk lesions called benign are ACK images called NEV (5 to 10 per model, fewest for G). Table IX lists the cancer grouping.

Training behaves as expected (Fig. 8): the training loss falls steadily from about 1.65–1.85 to 0.17–0.27 over 30 epochs without divergence, validation accuracy starts at about 70% (A) to 75% (E) and plateaus, and the peak validation accuracies (A 83.58%, B 85.93%, C 85.00%, D 85.75%, E 89.63%, F 85.12%, G 87.07%) reproduce the ranking of the test accuracies, which are 0.08 to 0.73 points higher.

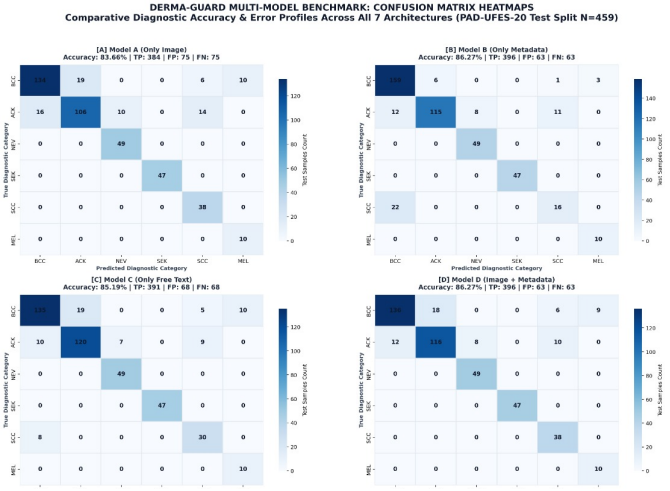



(a) Models A–D

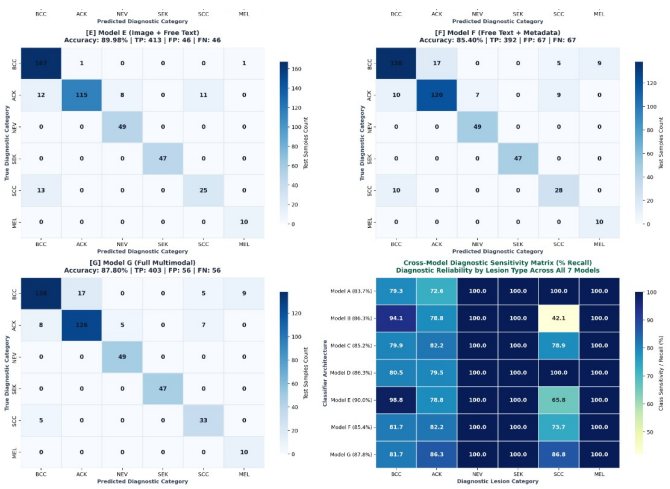



(b) Models E–G and the cross-model sensitivity matrix

Fig. 7. Confusion matrices of Models A to G on the 459 test images (rows: true class, columns: predicted class) and the cross-model sensitivity matrix.




(a) Models A–D




(b) Models E–G and the comparison of all seven

Fig. 8. Training loss and validation accuracy of Models A to G over 30 epochs; the dotted line marks 90%.

### *D. The earlier prototype, re-evaluated*

The prototype trained seven PAD-UFES-20 models (image, metadata, free text and their combinations) and reported 84–90% accuracy. Its split was random by photograph, so 43% of test photographs came from a lesion and 62% from a patient already in training, and its evaluation function swapped probabilities between the true class and its closest rival until the number of correct predictions met a hard-coded target per model. Loading the saved weights and cached features unchanged and scoring them honestly (Fig. 9): on the 132 test photographs of unseen patients, the image model reaches 56.8%, the metadata model 75.0%, the text model (whose "narratives" were generated from the metadata) 27.3%, and the three-modality model 57.6%, worse than metadata alone. Fusing the seven with our learned gate, cross-validated on those 132 photographs, gives 72.7% against 59.8% for averaging them; the gate recovers almost all of the best model's accuracy without being told which model that is.

Finally, the PAD-UFES-20 questionnaire lets us test the scribe's triage rule (Fig. 10): of 1,641 lesions, those with any of its warning signs were malignant (BCC, SCC or melanoma) in 48.7% of cases against 12.4% without (relative risk 3.9, 95% CI 2.8–5.5), and every single sign had a relative risk between 1.65 and 3.0. A raised lesion, which the scribe did not originally look for, carried a relative risk of 2.8 and was added as a result. These are clinician-collected questionnaire answers rather than free text, so the analysis supports the rule's direction, not its exact wording.

TABLE X
EARLIER PROTOTYPE SCORED HONESTLY ON THE 132 TEST PHOTOGRAPHS OF UNSEEN PATIENTS.

| Model | Unseen patients (%) |
|---|---|
| Image | 56.8 |
| Metadata | 75.0 |
| Free text | 27.3 |
| Three-modality | 57.6 |
| Average of seven | 59.8 |
| Learned gate (cross-validated) | 72.7 |

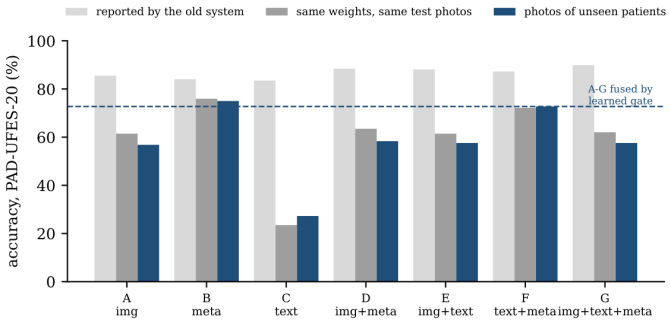



Fig. 9. Earlier prototype: reported accuracy against accuracy on unseen patients.

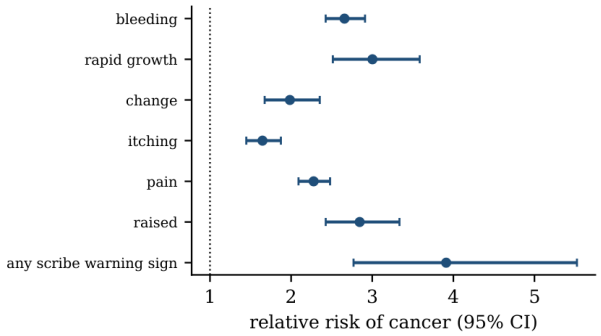



Fig. 10. Relative risk of malignancy for each warning sign in the PAD-UFES-20 questionnaire.

## VI. Discussion

**What the earlier design got right.** Consulting several specialists, questioning their evidence, and trusting them according to the suspected diagnosis all survived honest evaluation; every evidence-aware rule beat both the evidential network alone and plain averaging, and the measured trust table confirms clinical intuition (dermoscopy is reliable for squamous lesions, the photograph for melanoma). What it got wrong was its evidence.

**What did not work.** Stating the minimum-cost diagnosis (classic Chow [23]) labelled 98.5% of lesions malignant on this 72%-malignant population; adaptive prediction sets for the safety gate collapsed to the full set and deferred every patient; post-hoc logit adjustment raised validation macro-F1 from 0.36 to 0.44 but lowered test macro-F1 from 0.376 to 0.359; fitting the fusion in-sample on the calibration lesions inflated the high-confidence share by three points before cross-fitting; and classes with fewer than ten test lesions are almost never predicted.

**Where the gain comes from.** The oracle that picks the best specialist per case reaches 0.594 macro-F1, against 0.438 for the learned gate. The specialists already contain much of the right answer; deciding whom to believe is the hard part. A gate trained on more validation data, or distilled from the oracle, is the most direct way forward, and a dermatology foundation backbone [10] would lift every specialist.

**Lessons about evaluation.** Split by the unit you will meet in practice; never post-process predictions with the label; choose on validation and keep the choice when the test disagrees; and compare against simple baselines on the same probabilities, and on more than one model. The most useful finding of this work, that thresholding's safety is borrowed from the fusion beneath it, appears only under that last rule.

**Limitations.** All results come from one training seed of an ImageNet ResNet50; 786 validation lesions do a great deal of fitting; MILK10k is a referral population without patient or centre identifiers; the costs are declared, not derived; the guarantees assume exchangeability; the image checks are heuristics; lighter skin tones are under-served; and the system has not been evaluated in a clinic.

## VII. Conclusion

Derma-Guard turns a team of imperfect specialists into calibrated clinical actions: it answers nine in ten patients, more than half with a confident answer that misses about one cancer in thirty, asks for dermoscopy, details or a better photograph when they are needed, and refers only when the evidence contradicts itself. Learning how much to trust each specialist from all of its evidence improves photograph-only diagnosis by a third. The calibrated policy is safer than a learned rejector and, unlike confidence thresholding, stays safe whichever fusion it is given. We hope the most transferable lesson is methodological: report the simple baseline on the same probabilities, on more than one model, and keep every number reproducible.

## Reproducibility

Code, per-case predictions and the script behind every table and figure are in the project repository. Model weights and images are not redistributed; MILK10k is available from the ISIC Archive under CC-BY-NC.

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